

# \* CNN → Convolutional Neural Network

Why CNN?

↳ Traditional Neural Network struggle with images because images contain :-

- \* huge number of pixels.

- \* Spatial relationship

- \* patterns like edges, shapes, textures.

Eg → a simple image :-

$256 \times 256 \times 3$ .

Total Values :-

196,608 pixels.

→ If fully connected directly

- ↳ too many parameters.

- ↳ huge computation

- ↳ Overfitting.

CNN solves this problem.

Core idea of CNN

CNN tries to mimic how humans see patterns.

Instead of looking at the whole image at once.

- it looks at small regions

- detects simple features first.

- Gradually build complex understanding

Like :  $\rightarrow$

Edge  $\rightarrow$  Shapes  $\rightarrow$  Objects  $\rightarrow$  Full understanding  
@codingdidi

### CNN Architecture.

